

Question

Complex salt **A** contains 4 elements and exhibits higher conductivity than ordinary salts. It displays one-dimensional antiferromagnetism at low temperatures, and undergoes a phase transition to a paramagnetic structure as the temperature increases.

A chemical formula that reflects the structural characteristics of complex salt **A** is $[\text{BC}_2]\text{D}$, where: species **B** is a metallic element; species **C** is an electrically neutral ligand, and all elements within it have only one chemical environment; species **D** contains no metallic elements. Elemental analysis indicates that the mass fractions of *C* and *H* in **A** are 43.6% and 7.4%, respectively.

A is thermally unstable and must be prepared at very low temperatures ($< -40\text{ }^\circ\text{C}$). Vacuum decomposition of 1.54 g **A** at low temperature ($< 0\text{ }^\circ\text{C}$) yields 0.31 g of elemental **B**. All of this **B** is then reacted with O_2 under controlled conditions to obtain oxide **N**, with the chemical formula B_yO_z . All of the above **N** is dissolved in water, and 11.9 mL of a certain combustible gas is collected under standard conditions. The aforementioned aqueous solution is then titrated with 0.09196 mol/L HCl , consuming 25.32 mL.

Based on calculations, deduce the above unknowns and select the correct answer.

- A.** All other options are incorrect
- B.** The common valence states of **B** include +2.
- C.** 1.54 g **A** directly dissolved in sufficient sulfuric acid, the resulting gas is approximately 52 mL under standard conditions.
- D.** In the structure of **N**, the ratio of the number of chemical environments of **B** to the number of chemical environments of *O* is 2 : 1.
- E.** 1 mol **C** completely combusts, requiring the consumption of 18 mol O_2 .

F. The mass fraction of **D** in **A** is less than 0.1%

Answer

Correct Answer: **F**

Detailed Explanation

We can first determine the elements of **B**. The amount of OH^- produced by 0.31 g **B** is $n = c_{HCl}V_{HCl} = 0.09196 \text{ mol/L} \cdot 25.32 \text{ mL} = 0.00233 \text{ mol}$. Let $+a$ be the valence of **B** during titration, then the formula weight of **B** is $M_B = 0.31 \text{ g}/(0.00233 \text{ mol}/a) = 133.0/a \text{ g/mol}$

CHECKPOINT

3 PTS

Formula weight of **B** is $M_B = 133.0/a$

When $a = 1$, **B** is *Cs*; when $a = 2$, **B** is approximately *Zn*; when $a = 3$, **B** is approximately *Sc*. However, the hydroxides of the latter two are insoluble in water, so **B** is *Cs*.

CHECKPOINT

1 PTS

B is *Cs*

The H_2 produced by directly dissolving 0.31 g *Cs* in water is $V = (0.31 \text{ g}/133 \text{ g/mol})/2 \cdot 22.4 \text{ L/mol} \cdot 1000 \text{ mL/L} \approx 26.1 \text{ mL}$. According to the conservation of electrons in Cs_yO_z , we have $(y - 2z)/y = 11.9/26.1$, and the smallest reasonable integer solution is $y = 11, z = 3$. Therefore, **N** is $Cs_{11}O_3$. Since there is *Cs*(0) in it, **N** dissolves in water, and the flammable gas produced is the same as *Cs*, which is H_2 .

CHECKPOINT

3 PTS

N is $Cs_{11}O_3$

$M_A = M_{Cs} \cdot 1.54 \text{ g}/0.31 \text{ g} \approx 133 \text{ g/mol} \cdot 4.97 \approx 660.7 \text{ g/mol}$. According to the mass fraction of C and H , $n(C) = (660.7 \text{ g/mol} \cdot 0.436)/12.01 \text{ g/mol} \approx 23.98$, $n(H) = (660.7 \text{ g/mol} \cdot 0.074)/1.008 \text{ g/mol} \approx 48.50$.

Notice that $n(C) : n(H) \approx 1 : 2$, and **C** is a ligand for Cs^+ , and all elements have only one environment. According to coordination chemistry knowledge, it can be thought of as a crown ether ligand, and then **C** is deduced to be 18-crown-6 ($C_{12}H_{24}O_6$) based on the number of ligands. At this time, there is still $0.5H$ left, but the remaining molecular weight of **A** after deducting $Cs(C_{12}H_{24}O_6)_2$ is about 0. Considering the experimental error, combined with the fact that Cs has a +1 valence, **D** may be H^- or e^- . However, CsH will not decompose into Cs element at $< 0^\circ\text{C}$. Combining the information in the question, "the complex salt **A** contains 4 elements, has higher conductivity than ordinary salts, and has one-dimensional antiferromagnetism at low temperatures, while a phase transition occurs when the temperature rises, converting to a paramagnetic structure." This highly matches the properties of some electride salts, so **D** should be e^- , and **A** is the electride salt of Cs .

CHECKPOINT

2 PTS

C is 18-crown-6

CHECKPOINT

2 PTS

D is e^-

The common valence state of Cs is +1 instead of +2, so option B is incorrect.

CHECKPOINT

0.5 PTS

The common valence state of Cs is $+1$

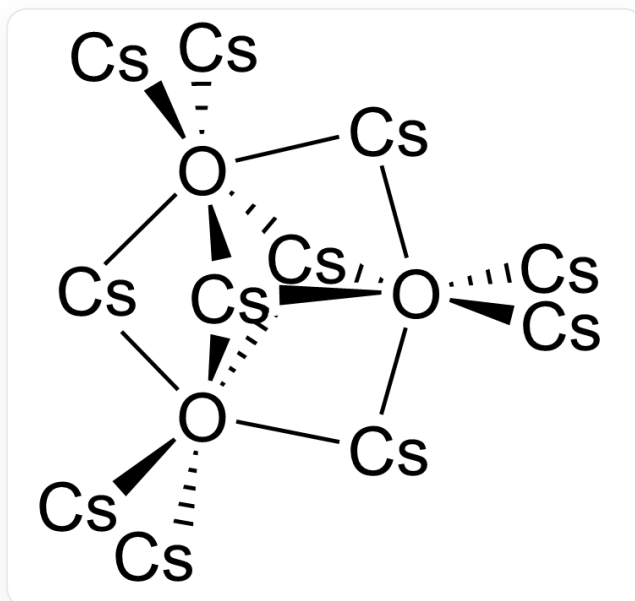
The amount of H_2 released by dissolving 1.54 g **A** in sulfuric acid can be regarded as the amount released by all 0.31 g Cs element contained in it, which is calculated as 26.1 mL H_2 as before, so option C is incorrect.

CHECKPOINT

1 PTS

1.54 g **A** releases 26.1 mL H_2 upon dissolution in sulfuric acid

The structure of $Cs_{11}O_3$ is shown in the figure below:



The cluster can be regarded as a triangle arrangement of three Cs_6O octahedra connected by shared faces. According to the different positions of cesium atoms in the cluster, there are three chemical environments: 3-coordinated, 2-coordinated, and 1-coordinated.

Therefore, there are 3 types of Cs and 1 type of O in $Cs_{11}O_3$, and the ratio is 3 : 1. Option D is incorrect.

CHECKPOINT

1.5 PTS

The ratio of the number of chemical environments of Cs to O in $Cs_{11}O_3$ is 3 : 1

Complete combustion of 1 mol 18-crown-6 ($C_{12}H_{24}O_6$) consumes $(12 + 24/4 - 6/2) = 15$ mol O_2 . Option E is incorrect.

CHECKPOINT

1 PTS

1 mol 18-crown-6 completely combusts consumes 15 mol O_2

The mass percentage of e^- in **A** is obviously $< 0.1\%$, so option F is correct.

CHECKPOINT

1 PTS

The mass percentage of e^- in **A** is $< 0.1\%$